

606029



Test Booklet No.

ENGLISH

Test Booklet Code

14

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below:
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen** only for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 14.** Make sure that the **CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet**. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

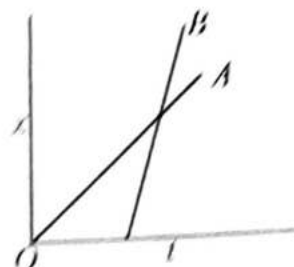
Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

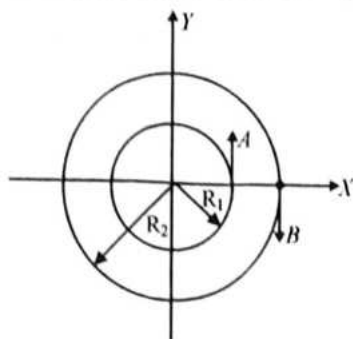
1. A proton and an α -particle are accelerated from rest by 2V and 4V potentials, respectively. The ratio of their de-Broglie wavelength is:
(1) 16 : 1 (2) 2 : 1
(3) 8 : 1 (4) 4 : 1
2. The resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is:
(1) 6.33 (2) 5.33
(3) 2.33 (4) 4.33
3. Time (T), velocity (C) and angular momentum (h) are chosen as fundamental quantities instead of mass, length and time, in terms of these, the dimensions of mass would be:
(1) $[M] = [T^{-1}C^2h]$ (2) $[M] = [T^1C^2h]$
(3) $[M] = [T^{-1}C^2h^{-1}]$ (4) $[M] = [TC^2h]$
4. The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is $x = 0$ at $t = 0$, then its displacement at time ($t = 1$) is:
(1) $v_0 + g + F$ (2) $v_0 + \frac{g}{2} + \frac{F}{3}$
(3) $v_0 + \frac{g}{2} + F$ (4) $v_0 + 2g + 3F$
5. The water drop falls at regular intervals from a tap 5 m above the ground. The third drop is leaving the tap at instant the first drop touches the ground. How far above the ground is the second drop at that instant?
(1) 3.75 m (2) 4.00 m
(3) 1.25 m (4) 2.50 m
6. The position-time graphs for two students A and B returning from the school to their homes are shown in figure.



- (A) A lives closer to the school
(B) B lives closer to the school
(C) A takes lesser time to reach home
(D) A travels faster than B
(E) B travels faster than A

Choose the correct answer from the options given below.

- (1) (A), (C) and (E) only
(2) (A), (C) and (E) only
(3) (A) and (E) only
(4) (B) and (E) only
7. A projectile is launched at an angle α with the horizontal with a velocity 20 ms^{-1} . After 10s, its inclination with horizontal is β . The value of $\tan \beta$ will be ($g = 10 \text{ ms}^{-2}$)
(1) $\tan \alpha + 5 \sec \alpha$ (2) $\tan \alpha - 5 \sec \alpha$
(3) $2 \tan \alpha - 5 \sec \alpha$ (4) $2 \tan \alpha - 5 \sec \alpha$
8. The coordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by:
(1) $3t\sqrt{\alpha^2 + \beta^2}$ (2) $3t^2\sqrt{\alpha^2 + \beta^2}$
(3) $t^2\sqrt{\alpha^2 + \beta^2}$ (4) $\sqrt{\alpha^2 + \beta^2}$
9. Two particles A and B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω . At $t = 0$, their positions and direction of motion are shown in the figure.



The relative velocity $\vec{v}_A - \vec{v}_B$ at $t = \frac{\pi}{2\omega}$ is given by:

- (1) $\omega(R_1 + R_2)\hat{i}$ (2) $-\omega(R_1 + R_2)\hat{i}$
 (3) $\omega(R_1 - R_2)\hat{i}$ (4) $\omega(R_2 - R_1)\hat{i}$

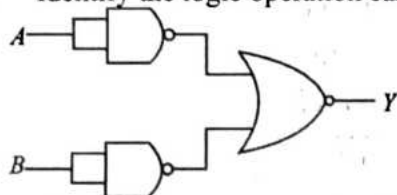
10. Three forces $F_1 = 10\text{N}$, $F_2 = 8\text{N}$, $F_3 = 6\text{N}$ are acting on particle of mass 5 kg. The force F_2 and F_3 are applied perpendicularly so that particle remains at rest. If the force F_1 is removed, then the acceleration of the particle is:

- (1) 0.5 ms^{-2} (2) 4.8 ms^{-2}
 (3) 2 ms^{-2} (4) 7 ms^{-2}

11. A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5s completely, the average force acting on that balloon will be (in dyne).

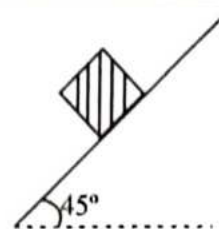
- (1) 3 (2) 9
 (3) 12 (4) 18

12. Identify the logic operation carried out.



- (1) OR (2) AND
 (3) NOR (4) NAND

13. Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down. The coefficient of friction between the block and inclined plane (μ) is equal to:



- (1) 0.50 (2) 0.25
 (3) 0.33 (4) 0.60

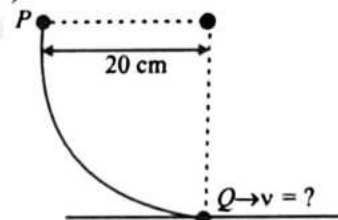
14. The potential energy of a 1 kg particle free to move along the x-axis is given by

$$V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \text{ J}$$

The total mechanical energy of the particle is 2J. Then, the maximum speed (in ms^{-1}) is:

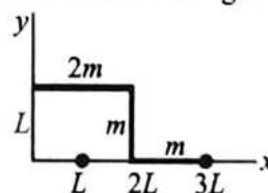
- (1) $3/\sqrt{2}$ (2) $\sqrt{2}$
 (3) $1/\sqrt{2}$ (4) 2

15. As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be ($g = 10 \text{ m/s}^2$)



- (1) 4 m/s (2) 0
 (3) 2 m/s (4) 0.25 m/s

16. The position vector of the centre of mass r_{cm} of an asymmetric uniform bar of negligible area of cross-section as shown in figure is:



$$(1) r = \frac{13}{8}Lx + \frac{5}{8}Ly \quad (2) r = \frac{11}{8}Lx + \frac{3}{8}Ly$$

$$(3) r = \frac{3}{8}Lx + \frac{11}{8}Ly \quad (4) r = \frac{5}{8}Lx + \frac{13}{8}Ly$$

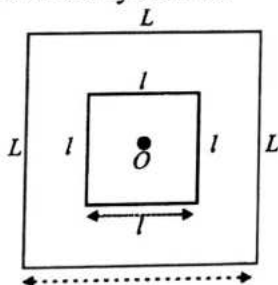
17. Match List-I with List-II.

List-I	List-II
(A) Moment of inertia of solid sphere of radius R about any tangent.	(I) $\frac{5}{3}MR^2$
(B) Moment of inertia of hollow sphere of radius (R) about any tangent.	(II) $\frac{7}{5}MR^2$
(C) Moment of inertia of circular ring of radius (R) about its diameter.	(III) $\frac{1}{4}MR^2$
(D) Moment of inertia of circular disc of radius (R) about any diameter.	(IV) $\frac{1}{2}MR^2$

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
 (2) A-I, B-II, C-IV, D-III
 (3) A-II, B-I, C-III, D-IV
 (4) A-I, B-II, C-III, D-IV

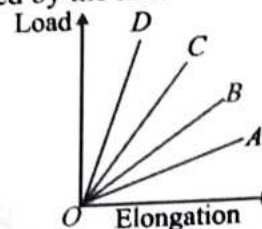
18. A small square loop of wire of side l is placed inside a large square loop of wire $L (L \gg l)$. Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is:



$$(1) \frac{2\sqrt{2}\mu L^2}{\pi l} \quad (2) \frac{\mu_0 l^2}{2\sqrt{2}\pi L}$$

$$(3) \frac{2\sqrt{2}\mu_0 l^2}{\pi L} \quad (4) \frac{\mu_0 L^2}{2\sqrt{2}\pi l}$$

19. The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line.

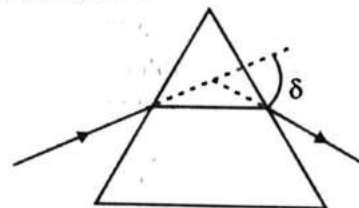


- (1) OA
 (2) OC
 (3) OD
 (4) OB

20. A light cylindrical vessel is kept on a horizontal surface. Area of base is A . A hole of cross-sectional area a is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is ($a \ll A$).

- (1) $\frac{A}{2a}$
 (2) $\frac{2A}{a}$
 (3) $\frac{2a}{A}$
 (4) $\frac{a}{A}$

21. The angle of deviation through a prism is minimum when



- A. incident ray and emergent ray are symmetric to the prism
 B. the refracted ray inside the prism becomes parallel to its base
 C. angle of incidence is equal to that of the angle of emergence
 D. angle of emergence is double the angle of incidence

Choose the **correct** answer from the options given below.

- (1) Statements (A), (B) and (C) are true.
- (2) Only statement (D) is true.
- (3) Only statements (A) and (B) are true.
- (4) Statements (B) and (C) are true.

22. Assume that a drop, of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T , density of liquid is ρ and L is its latent heat of vaporization.

- (1) $\frac{2T}{\rho L}$
- (2) $\frac{\rho L}{T}$
- (3) $\sqrt{\frac{T}{\rho L}}$
- (4) $\frac{T}{\rho L}$

23. n moles of an ideal gas with constant volume heat capacity C_V undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is:

- (1) $\frac{4nR}{C_V + nR}$
- (2) $\frac{4nR}{C_V - nR}$
- (3) $\frac{nR}{C_V - nR}$
- (4) $\frac{nR}{C_V + nR}$

24. Consider a sample of oxygen behaving like an ideal gas. At 300 K, the ratio of root mean square (rms) velocity to the average velocity of gas molecule would be (Molecular weight of oxygen is 32 g/mol; $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$)

- (1) $\sqrt{\frac{3}{3}}$
- (2) $\sqrt{\frac{8}{3}}$
- (3) $\sqrt{\frac{3\pi}{8}}$
- (4) $\sqrt{\frac{8\pi}{3}}$

25. The power factor of an AC circuit having resistance R and inductance L (connected in series) and an angular velocity ω is:

- (1) $\frac{R}{\omega L}$
- (2) $\frac{R}{(R^2 + \omega^2 L^2)^{1/2}}$
- (3) $\frac{\omega L}{R}$
- (4) $\frac{R}{(R^2 - \omega^2 L^2)^{1/2}}$

26. A particle of charge q and mass m is moving with a velocity $-vi$ ($v \neq 0$) towards a large screen placed in the yz -plane at a distance d . If there is a magnetic field $B = B_0 k$, the maximum value of v for which the particle will not hit the screen is:

- (1) $\frac{qdB_0}{3m}$
- (2) $\frac{2qdB_0}{m}$
- (3) $\frac{qdB_0}{m}$
- (4) $\frac{qdB_0}{2m}$

27. Sixty four conducting drops each of radius 0.02 m and each carrying a charge of $5 \mu\text{C}$ are combined to form a bigger drop. The ratio of surface charge density of bigger drop to the smaller drop will be:

- (1) 1 : 4
- (2) 4 : 1
- (3) 1 : 8
- (4) 8 : 1

28. The electric field due to a short electric dipole at a large distance (r) from centre of dipole on the equatorial plane varies with distance as

- (1) $\frac{1}{r^2}$
- (2) $\frac{1}{r^3}$
- (3) $\frac{1}{r}$
- (4) r

29. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1)/ V_2 (on S_2) of the electrostatic potentials on each sphere is:

- (1) $\left(\frac{R_1}{R_2}\right)^3$
- (2) $\frac{R_2}{R_1}$
- (3) $\frac{R_1}{R_2}$
- (4) $\left(\frac{R_1}{R_2}\right)^2$

30. To light a 50 W, 100 V lamp is connected in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}} \mu F$ 200 V, 50Hz Ac source. The value of x will be.

(1) 1 (2) 4
(3) 3 (4) 2

31. A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$, where $\alpha_0 = 20 \text{ A/s}$ and $\beta = 8 \text{ As}^{-2}$. Find the charge crossed through a section of the wire in 15 s.

(1) 260 C (2) 2100 C
(3) 11250 C (4) 2250 C

32. In a Wheatstone's bridge, three resistances P , Q and R are connected in the three arms and the fourth arm is formed by two resistances S_1 and S_2 connected in parallel. The condition for the bridge to be balanced will be:

(1) $\frac{P}{Q} = \frac{2R}{S_1 + S_2}$ (2) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{S_1 S_2}$
(3) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{2S_1 S_2}$ (4) $\frac{P}{Q} = \frac{R}{S_1 + S_2}$

33. Two identical wires A and B , each of length l , carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side a . If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is:

(1) $\frac{\pi^2}{8}$ (2) $\frac{\pi^2}{16\sqrt{2}}$
(3) $\frac{\pi^2}{16}$ (4) $\frac{\pi^2}{8\sqrt{2}}$

34. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

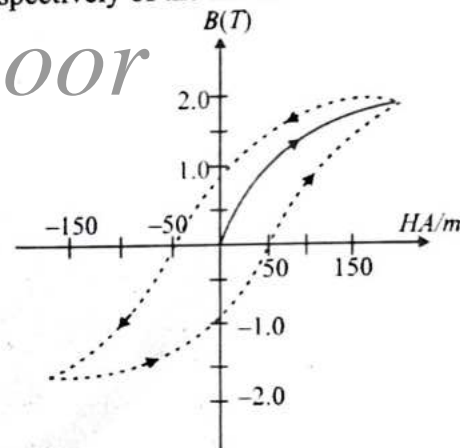
Assertion (A): The phase difference of two light waves changes if they travel through different media having same thickness, but different indices of refraction.

Reason (R): Phase difference between two waves do not depend on wavelengths of two waves.

In the light of the above statements, choose the correct answer from the options given below.

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

35. The figure gives experimentally measured B versus H variation in a ferromagnetic material. The retentivity, coercivity and saturation respectively of the material are:



- (1) 1.0 T, 50 A/m and 1.5 T
(2) 150 A/m, 1.0 T and 1.5 T
(3) 1.5 T, 50 A/m and 1.0 T
(4) 1.5 T, 50 A/m and 1.0 T

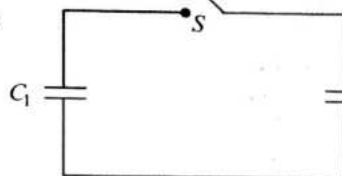
Physics : Section-B (Q. No. 36 to 50)

36. A wire of length 1 m moving with velocity 8 m/s at right angles to a magnetic field of 2T. The magnitude of induced emf between the ends of wire will be
- (1) 8 V (2) 12 V
(3) 16 V (4) 20 V

37. The angular momentum of a planet of mass M moving around the Sun in an elliptical orbit is L . The magnitude of the areal velocity of the planet is:

(1) $\frac{4L}{M}$ (2) $\frac{L}{M}$
 (3) $\frac{2L}{M}$ (4) $\frac{L}{2M}$

38. Two capacitors having capacitance C_1 and C_2 , respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S . The amount of charge on the capacitor C_2 , after equilibrium is:



(1) $\frac{C_1 C_2}{(C_1 + C_2)} V$ (2) $\frac{(C_1 + C_2)}{C_1 C_2} V$
 (3) $(C_1 + C_2) V$ (4) $(C_1 - C_2) V$

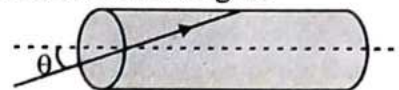
39. A particle starts executing simple harmonic motion (SHM) of amplitude a and total energy E . At any instant, its kinetic energy is $3E/4$, then its displacement y is given by.

(1) $y = a$ (2) $y = \frac{a}{\sqrt{2}}$
 (3) $y = \frac{a\sqrt{3}}{2}$ (4) $y = \frac{a}{2}$

40. Light wave travelling in air along x -direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct) \text{ Vm}^{-1}$. Then the peak value of magnetic field of wave will be (Given, $c = 3 \times 10^8 \text{ ms}^{-1}$)

(1) $18 \times 10^{-7} \text{ T}$ (2) $54 \times 10^{-7} \text{ T}$
 (3) $54 \times 10^{-8} \text{ T}$ (4) $18 \times 10^{-8} \text{ T}$

41. A transparent solid cylinder rod has a refractive index of $2/\sqrt{3}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.

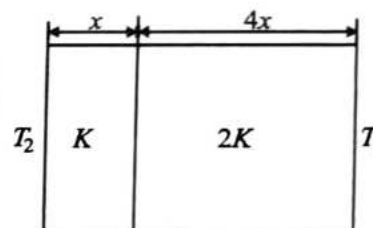


The incident angle θ for which the light ray grazes along the wall of the rod is

(1) $\sin^{-1}\left(\frac{1}{2}\right)$ (2) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
 (3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

42. The temperature of two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$ respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right) f$, then

f is equal to



(1) 1 (2) 1/2
 (3) 2/3 (4) 1/3

43. Given below are two statements:

Statement-I: If the Brewster's angle of the light propagating from air to glass is θ_B , then the Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2} - \theta_B$.

Statement-II: The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_g)$, where μ_g is the refractive index of glass.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

44. A metallic surface is illuminated with radiation of wavelength λ , the stopping potential is V_0 . If the same surface is illuminated with radiation of wavelength 2λ , the stopping potential becomes $V_0/4$. The threshold wavelength for this metallic surface will be:

- (1) 3λ
- (2) $\frac{3}{2}\lambda$
- (3) $\frac{\lambda}{4}$
- (4) 4λ

45. Velocity (v) and acceleration (a) in the systems of unit 1 and 2 are related as $v_2 = \frac{n}{m^2} v_1$ and

$a_2 = \frac{a_1}{mn}$, respectively. Here, m and n are

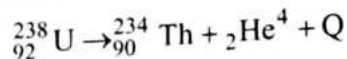
constants. The relations for distance and time in two systems respectively are:

- (1) $\frac{n^3}{m^3} L_1 = L_2$ and $\frac{n^2}{m} T_1 = T_2$
- (2) $L_1 = \frac{n^4}{m^2} L_2$ and $T_1 = \frac{n^2}{m} T_2$
- (3) $L_1 = \frac{n^4}{m} L_2$ and $T_1 = \frac{n^2}{m^2} T_2$
- (4) $\frac{n^2}{m} L_2 = L_1$ and $\frac{n^4}{m^2} T_1 = T_2$

46. The angular momentum for the electron in Bohr's orbit is L . If the electron is assumed to revolve in second orbit of hydrogen atom, then the change in angular momentum will be:

- (1) L
- (2) $L/2$
- (3) Zero
- (4) $2L$

47. A common example of a alpha decay is



Given,

$${}_{92}^{238}\text{U} = 238.05060 \text{ u}, {}_{90}^{234}\text{Th} = 234.04360 \text{ u},$$

$${}_2^4\text{He} = 4.00260 \text{ u} \text{ and } 1 \text{ u} = 931.5 \frac{\text{MeV}}{c^2}$$

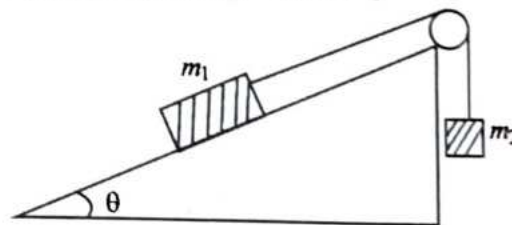
The energy released (Q) during the alpha decay of ${}_{92}^{238}\text{U}$ is MeV

- (1) 1 MeV
- (2) 3 MeV
- (3) 6 MeV
- (4) 4 MeV

48. A ray is incident at an angle of incidence i on one surface of a prism of small angle A and emerge normally from opposite surface. If the refractive index of the material of prism is μ , the angle of incidence i is nearly equal to:

- (1) $\frac{A}{\mu}$
- (2) $\frac{A}{2\mu}$
- (3) μA
- (4) $\frac{\mu A}{2}$

49. The bodies of masses $m_1 = 5 \text{ kg}$ and $m_2 = 3 \text{ kg}$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be [Take, $g = 10 \text{ ms}^{-2}$]

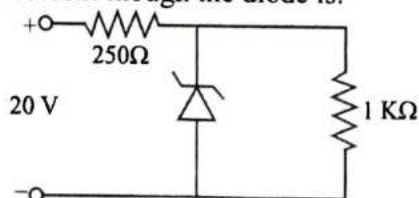


- (1) 30 N
- (2) 40 N
- (3) 10 N
- (4) 20 N

50. A Zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure.



The current through the diode is:



- (1) 10 mA (2) 15 mA
(3) 20 mA (4) 5 mA

Chemistry : Section-A (Q. No. 51 to 85)

51. In the reaction;
 $\text{MnO}_4^- + \text{SO}_3^{2-} + \text{H}^+ \rightarrow \text{SO}_4^{2-} + \text{Mn}^{2+} + \text{H}_2\text{O}$
 (1) MnO_4^- and H^+ both are reduced
 (2) MnO_4^- is reduced and H^+ is oxidised
 (3) MnO_4^- is reduced and SO_3^{2-} is oxidised
 (4) MnO_4^- is oxidised and SO_3^{2-} is reduced
52. Which condition favours the bond formation:
 (1) Maximum attraction and maximum potential energy
 (2) Minimum attraction and minimum potential energy
 (3) Minimum potential energy and maximum attraction
 (4) Minimum potential energy and minimum attraction
53. 1.6 g SO_2 and 1.5×10^{22} Molecules of H_2S are mixed and allowed to react completely in closed vessel as follows;
 $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
 Which of the following statement is/are incorrect?
 (A) 0.025 moles H_2S present initially in reaction Vessel
 (B) 0.0125 moles H_2S present after the completion of reaction
 (C) 0.0125 moles of SO_2 reacts with H_2S
 (D) 0.025 moles of SO_2 reacts with H_2S
 (1) Only (A) (2) Both (B) and (D)
 (3) (A), (B) and (C) (4) (A), (C) and (D)
54. Which of the following is a poor conductor of electricity?
 (1) CH_3COONa (2) $\text{C}_2\text{H}_5\text{OH}$
 (3) NaCl (4) KOH

55. Given below are two statements.

Statement I: One atomic mass unit is defined as one-twelfth of the mass of one carbon-14 atom.

Statement II: The carbon-14 isotope is the most abundant isotope of carbon and has been chosen as the standard.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Statement-I is correct but Statement-II is incorrect.
 (2) Statement-I is incorrect but Statement-II is correct.
 (3) Both Statement-I and Statement-II are correct.
 (4) Both Statement-I and Statement-II are incorrect.

56. Number of radial nodes are same for which of the following orbitals?

- A. 4d B. 2s
 C. 3s D. 5f
 E. 2p

Choose the most appropriate option from the following.

- (1) (C), (D) and (E)
 (2) (A) and (B)
 (3) (B), (C), (D) and (E)
 (4) (A), (B) and (D)

57. The compound A undergoes 100% tetramerization in a given organic solvent. The van't Hoff factor i is;

- (1) 4.0 (2) 0.25
 (3) 0.125 (4) 2.0

58. On heating an aliphatic primary amine with chloroform and ethanolic KOH , the organic compound formed is:

- (1) $\text{R}-\text{CH}_2-\text{CN}$ (2) $\text{R}-\text{CH}_2\text{NC}$
 (3) $\text{R}-\text{CH}_2-\text{OH}$ (4) $\text{R}-\text{CH}_2-\text{CHO}$

59. Given below are two statements.

Statement I: Sucrose is dextrorotatory but after hydrolysis Solution becomes Laevorotary.

Statement II: Laevorotation of fructose (-92.4°) is more than dextrorotation of glucose ($+52.5^\circ$). In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

60. The **correct** order of reactivity of PhMgBr with the following is;

- (1) Benzophenone > Acetophenone > Acetone
- (2) Benzophenone < Acetophenone < Acetone
- (3) Acetophenone < Benzophenone < Acetone
- (4) Acetophenone > Benzophenone > Acetone

61. Which one of the following will have highest basicity in gaseous medium?

- (1) $\text{C}_2\text{H}_5\text{NH}_2$
- (2) $(\text{C}_2\text{H}_5)_3\text{N}$
- (3) $(\text{C}_2\text{H}_5)_2\text{NH}$
- (4) NH_3

62. Match List-I with List-II.

List-I
(Compound)

List-II
(Shape)

- | | |
|------------------------|----------------------------|
| (A) $[\text{ICl}_4]^-$ | (I) Pentagonal bipyramidal |
| (B) ClF_5 | (II) Pyramidal |
| (C) XeO_3 | (III) Square planar |
| (D) IF_7 | (IV) Square pyramidal |

Choose the **correct** answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

63. The standard enthalpy of formation is non-zero for:

- (1) $\text{Br}_2(l)$
- (2) S(rhombic)
- (3) C(graphite)
- (4) C(diamond)

64. Which of the following is **not** an interhalogen compound?

- (1) ClF
- (2) ClF_3
- (3) HF
- (4) IF_5

65. Match List-I with List-II.

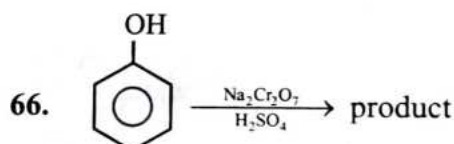
List-I
(Reduction Reaction)

List-II
(Catalyst)

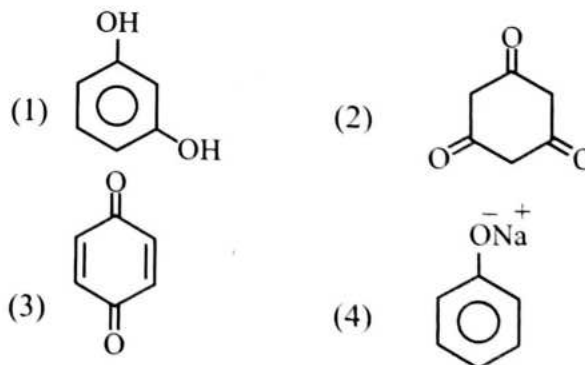
- | | |
|-----------------------------|---|
| (A) Stephen reduction | (I) $\text{H}_2/\text{Pd}-\text{BaSO}_4$ |
| (B) Clemmensen's reduction | (II) $\text{N}_2\text{H}_4/\text{KOH}/\text{HOCH}_2\text{CH}_2-\text{OH}$ |
| (C) Rosenmund's reduction | (III) SnCl_2/HCl |
| (D) Wolff-Kishner reduction | (IV) $\text{Zn}-\text{Hg}/\text{Conc. HCl}$ |

Choose the **correct** answer from the options given below.

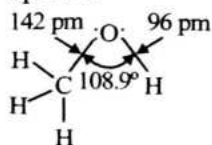
- (1) A-III, B-IV, C-I, D-II
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-III, C-II, D-I



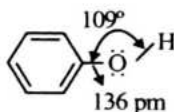
The product is;



67. From the following compounds, choose **correct** options



Methanol



Phenol

- (A) Phenol is less acidic than methanol due to hydrogen bonding in phenol
 (B) Bond angle in phenol is more than methanol
 (C) C-O bond length in phenol is less than in alcohol as phenol has partial double bond character due to conjugation

Choose the *most appropriate* answer from the options given below.

- (1) A and B only
 (2) A and C only
 (3) B and C only
 (4) A, B and C

68. Match List-I with List-II.

List-I

(Basic radicals)

- (A) NH_4^+
 (B) Zn^{2+}

- (C) Ba^{2+}

- (D) Pb^{2+}

List-II

(Group reagents)

- (I) Dil HCl
 (II) $(\text{NH}_4)_2\text{CO}_3$ Solution in presence of NH_4Cl and NH_4OH
 (III) Concentrated NaOH solution
 (IV) $\text{H}_2\text{S}(\text{g})$ in presence of $\text{NH}_4\text{Cl}(\text{s})$ and NH_4OH (excess)

Choose the **correct** answer from the options given below.

- (1) A-I, B-II, C-III, D-IV
 (2) A-II, B-IV, C-I, D-III
 (3) A-III, B-IV, C-II, D-I
 (4) A-IV, B-III, C-I, D-II

69. Equal volumes of 0.2 M HCl and 0.1 M NaOH are mixed. The concentration of the resulting solution is:

- (1) 0.1 M (2) 0.05 M
 (3) 0.2 M (4) 0.02 M

70. The Vitamins that can be stored in our body is:

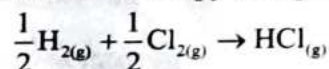
- A. Vitamin A B. Vitamin B_1
 C. Vitamin B_6 D. Vitamin B_{12}
 E. Vitamin C F. Vitamin D
 G. Vitamin E

- (1) (A), (C), (G)
 (2) (B), (C), (E)
 (3) (D), (E), (F), and (G)
 (4) (A), (D), (F) and (G)

71. Amphoteric oxide among the following is;

- (1) Mn_2O_7 (2) V_2O_5
 (3) CrO_3 (4) CrO

72. The standard enthalpy of formation of $\text{H}_{2(\text{g})}$ and $\text{Cl}_{2(\text{g})}$ and $\text{HCl}_{(\text{g})}$ are 218 kJ/mol, 121.68 kJ/mol and -92.3 kJ/mol respectively. Calculate standard enthalpy change in kJ for;



- (1) +431.99 (2) -262.14
 (3) -431.99 (4) +247.37

73. Be^{2+} is isoelectronic with which of the following ions?

- (1) H^+ (2) Li^+
 (3) Na^+ (4) Mg^{2+}

74. Select the molecule which has only one π -bond?

- (1) $\text{CH} \equiv \text{CH}$
 (2) $\text{CH}_2 = \text{CH} - \text{CHO}$
 (3) $\text{CH}_3 - \text{CH} = \text{CH}_2$
 (4) $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$

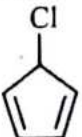
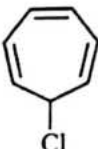
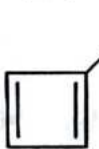
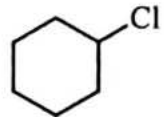
75. The first order reaction is completed in 60 minutes, then 50% reaction completes approximately in;

- (1) 40 min (2) 45 min
 (3) 60 min (4) 50 min

76. The molar depression constant of water is $1.86^\circ\text{C mole}^{-1}$. If 342 gm of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) are dissolved in 1000 gm of water, the solution will freeze at;

- (1) -1.86°C (2) 1.86°C
 (3) -3.92°C (4) 2.42°C

77. When 3 mole of A and 1 mole of B are mixed in 1 litre vessel the following reaction takes place
 $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$
 1.5 moles of C are formed till equilibrium is established. The equilibrium constant for the reaction is
 (1) 0.12 (2) 0.25
 (3) 0.50 (4) 4.0
78. From the following element choose the **correct** options
 The general electronic configuration of f-block elements is
 $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$
 (A) General oxidation state of Actinoids is +3
 (B) Pr and Nd exhibit +4 oxidation state in oxides only
 (C) Th (thorium) can show +6 oxidation state
 Choose the *most appropriate* answer from the options given below.
 (1) A and B only (2) A, B and C
 (3) A only (4) A and C only
79. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): On increasing temperature, rate of reaction increase.
Reason (R): Fraction of molecules, which collide with energies greater and equal than E_a increases.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.
80. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.8g of the compound, which neutralized 20mL of 1M H_2SO_4 . The percentage of nitrogen in the compound is;
 (1) 40% (2) 60%
 (3) 70% (4) 80%

81. Given below are two statements.
Statement I: One moles of HBr on reaction with ethene forms monobromo substituted product.
Statement II: Two moles of HBr on reaction with propyne forms 2,2-Dibromopropane.
 In the light of the above statements, choose the *most appropriate* answer from the options given below.
 (1) Statement I and Statement II both are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Statement I and Statement II both are incorrect.
82. Which of the following groups of elements is assigned zero electronegativity?
 (1) Alkali metals
 (2) Noble gases
 (3) Alkaline earth metals
 (4) Rare earth metals
83. Which compound will show fastest reaction with $AgNO_3$?
 (1)  (2) 
 (3)  (4) 
84. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Electron gain enthalpy of Fluorine is less negative than chlorine.
Reason (R): Fluorine is small in size therefore have strong inter electronic repulsion.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

85. From the following Statements choose the correct options

The effect of temperature on spontaneity of reaction is observed for endothermic and exothermic.

- (A) Exothermic reactions are spontaneous at high temperature
(B) Endothermic reactions are Spontaneous at high temperature
(C) Endothermic reactions are spontaneous at low temperature

Choose the most appropriate answer from the options given below.

- (1) A only (2) B only
(3) A and B only (4) A, B and C

Chemistry : Section-B (Q. No. 86 to 100)

86. Match List-I with List-II.

List-I (Complex)	List-II (CFSE)
(A) $\text{Cu}(\text{NH}_3)_6^{2+}$	(I) $-0.6 \Delta_o$
(B) $\text{Ti}(\text{H}_2\text{O})_6^{3+}$	(II) $-2.0 \Delta_o$
(C) $\text{Fe}(\text{CN})_6^{3+}$	(III) $-1.2 \Delta_o$
(D) $[\text{NiF}_6]^{4-}$	(IV) $-0.4 \Delta_o$

Choose the correct answer from the options given below.

- (1) A-I, B-IV, C-II, D-III
(2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

87. The correct statement for the reaction is;
 $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$

- (1) Hybridisation of N changes
(2) Bond angle increases
(3) NH_3 act as a Lewis acid
(4) Regular geometry is changed

88. $2\text{CH}_3 - \text{CHO} \xrightarrow{\text{(i) dil NaOH}} [\text{A}] \xrightarrow{\Delta} [\text{B}]$

The compound A and B formed respectively are;

- (1) $\text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$ and $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$
(2) $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2 - \text{CHO}$ and $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CHO}$
(3) $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2 - \text{CHO}$ and $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$
(4) $\text{CH}_3 - \text{CH}_2 - \text{CH}(\text{OH}) - \text{CHO}$ and $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$

89. $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow{\text{(i) BD}_3} \text{A, A is:}$
 $\xrightarrow{\text{(ii) H}_2\text{O}_2/\text{OH}^-}$

- (1) $\text{CH}_3 - \underset{\text{D}}{\text{CH}} - \text{CH}_2 - \text{OH}$
(2) $\text{CH}_3 - \underset{\text{O}}{\underset{\text{||}}{\text{CH}}} - \text{CH}_2 - \text{D}$
(3) $\text{CH}_3 - \underset{\text{D}}{\text{CH}} - \text{CHO}$
(4) $\text{CH}_3 - \underset{\text{D}}{\text{CH}} - \underset{\text{O}}{\underset{\text{||}}{\text{C}}} - \text{D}$

90. On the basis of Le-chatelier principle in equilibrium

- A. Addition of inert gas, at constant volume do not disturb equilibrium
B. Addition of inert gas, at constant pressure do not disturb equilibrium
C. Addition of catalyst do not disturb the equilibrium

Choose the most appropriate answer from the options given below.

- (1) A, B and C (2) A and B only
(3) A and C only (4) B and C only

91. The hydrogen electrode is dipped in a solution of pH = 3 at 25° C. The potential of the cell would be (the value of $2.303 RT/F$ is 0.059 V)

- (1) 0.177 V (2) -0.177 V
(3) 0.087 V (4) 0.059 V

92. Which compound is used in manufacturing of $\text{K}_2\text{Cr}_2\text{O}_7$?

- (1) Na_2CrO_4 (2) CrO_5
(3) FeCr_2O_4 (4) CrO_3

93. An alcohol ($\text{C}_2\text{H}_5\text{OH}$) reacts with conc. H_2SO_4 acid at 443K form final product is;

- (1) $\text{CH}_3 - \text{CH}_3$ (2) $\text{CH}_3 - \overset{\oplus}{\text{CH}}_2$
(3) $\text{CH}_3 - \text{CH}_2 - \overset{\oplus}{\text{O}}\text{H}$ (4) $\text{CH}_2 = \text{CH}_2$

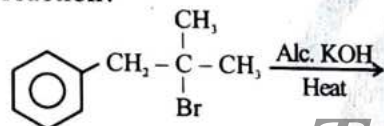
94. Which of the following is the coordination entity in $\text{K}_2[\text{Zn}(\text{OH})_4]$?

- (1) K^+ (2) Zn^{2+}
(3) OH^- (4) $[\text{Zn}(\text{OH})_4]^{2-}$

95. Which of the following statements regarding white phosphorus are correct?
 A. White phosphorus is more reactive than red phosphorus
 B. White phosphorus is non poisonous
 C. White phosphorus can dissolve in boiling NaOH
 Choose the *most appropriate* answer from the options given below.
 (1) A and B only (2) A, B and C
 (3) A and C only (4) B and C only

96. The rate of disappearance of SO_2 in the reaction $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ is $1.28 \times 10^{-3} \text{ g/s}$ then the rate of formation of SO_3 is
 (1) $0.64 \times 10^{-3} \text{ g/s}$ (2) $0.8 \times 10^{-3} \text{ g/s}$
 (3) $1.285 \times 10^{-3} \text{ g/s}$ (4) $1.6 \times 10^{-3} \text{ g/s}$

97. What is the major product of the following reaction?



- (1)
 (2)
 (3)
 (4)

98. Which one of the following compound is most reactive for E_1 reaction?

- (1) $\text{CH}_3-\text{CH}_2-\text{C}(\text{Br})(\text{CH}_3)-\text{CH}_3$ (2) $\text{CH}_3-\text{CH}_2-\text{C}(\text{Br})(\text{CH}_3)-\text{C}_6\text{H}_5$
 (3) $\text{CH}_3-\text{CH}_2-\text{C}(\text{Br})(\text{CH}_3)-\text{C}_6\text{H}_5$ (4) $\text{CH}_3-\text{CH}_2-\text{CH}(\text{Br})-\text{CH}_3$

99. Which one of the following resonating structure is most stable?

- (1) $\text{CH}_2=\text{CH}-\text{CH}=\text{O}$
 (2) $\overset{\ominus}{\text{C}}\text{H}_2=\overset{\oplus}{\text{C}}\text{H}-\text{CH}=\text{O}$
 (3) $\text{CH}_2=\text{CH}-\overset{\oplus}{\text{C}}\text{H}-\overset{\ominus}{\text{O}}$
 (4) $\overset{\oplus}{\text{C}}\text{H}_2-\text{CH}=\text{CH}-\overset{\ominus}{\text{O}}$

100. less basic than due to;

- (1) -I of $-\text{CH}_3$
 (2) +I of $-\text{CH}_3$
 (3) +H-effect of $-\text{CH}_3$
 (4) steric inhibition of protonation by $-\text{CH}_3$

Botany : Section-A (Q. No. 101 to 135)

101. A heterozygous pea plant with violet flowers was crossed with homozygous pea plant with white flower. Violet is dominant over white. Which one of the following represents the expected combinations among 40 progenies formed?

- (1) 30 produced violet and 10 produced white flowers
 (2) 20 produced violet and 20 produced white flowers
 (3) All 40 produced violet flowers
 (4) All 40 produced white flowers

102. An example of *ex-situ* conservation is:

- (1) wildlife sanctuaries.
 (2) biosphere reserves.
 (3) seed bank.
 (4) sacred groves.

103. Which among the following options is **incorrect** w.r.t. the class of pteridophytes and its respective example?

- (1) Psilopsida – *Psilotum*
 (2) Lycopsidea – *Selaginella*
 (3) Sphenopsida – *Adiantum*
 (4) Pteropsida – *Dryopteris*

104. Which of the following is transcribed by RNA polymerase III?

- (a) tRNA (b) 28S rRNA
(c) 5.8S rRNA (d) snRNA
(e) hnRNA (f) 5SrRNA
(1) (b), (c), (d) (2) (a), (d), (f)
(3) (c), (d), (e) (4) (a), (b), (c)

105. Which of the following group of organisms serve as an important biofertilizer in paddy fields?

- (1) Methanogens (2) Cyanobacteria
(3) Archaeobacteria (4) Protozoans

106. Given below are two statements:

Statement-I: In anaerobic organisms, glycolysis is the only process in respiration.

Statement-II: Glycolysis occurs in some living organisms.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

107. During decomposition of detritus, bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called;

- (1) fragmentation. (2) leaching.
(3) humification. (4) catabolism.

108. Select the **incorrect** statement w.r.t. Calvin cycle.

- (1) Carboxylation is the most crucial step of the Calvin cycle where CO_2 is utilised for the carboxylation of RuBP.
(2) Carboxylation is catalysed by the enzyme RuBP carboxylase.
(3) Reduction in Calvin cycle is a series of reactions that lead to the formation of 3-PGA.
(4) The regeneration steps require one ATP for phosphorylation to form RuBP.

109. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The trichomes help in preventing water loss due to transpiration.

Reason (R): The trichomes in the shoot system are usually multicellular.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is *not* the correct explanation of A.

110. Consider the two statements.

A. Plots with more species showed less year-to-year variation in total biomass.

B. Increased diversity contributed to higher productivity.

These statements are **true** regarding:

- (1) The evil quartet.
(2) David Tilman's long-term ecosystem experiments.
(3) The Earth Summit.
(4) Species-area relationships studied by Alexander von Humboldt.

111. Match List-I with List-II.

List-I (causal organism)	List-II (disease)
A. Virus	I. Mosaic disease
B. Plasmodium	II. Malaria
C. Bacteria	III. Cholera
D. Viroids	IV. Potato spindle tuber disease

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

112. Following are the statements about *Bacillus thuringiensis*.

- Bacteria *Bacillus thuringiensis* can control earthworm caterpillars.
- In the gut, Bt toxin is released and kill the caterpillars and leave other insects unharmed.
- The bacteria *Bacillus thuringiensis* is a microbial biocontrol agent.
- The scientists are able to introduce Bt toxin genes in to plants, in order to produce pest resistant plants.

Select the option with all correct statements:

- (1) (a) & (b) (2) (b), (c) & (d)
- (3) (a), (b) & (d) (4) (a), (b) & (c)

113. Which family exhibits actinomorphic, pedicellate, bisexual, complete, pentamerous flowers, typically protected by an epicalyx?

- (1) Malvaceae (2) Cruciferae
- (3) Compositae (4) Gramineae

114. Match List-I with List-II.

List-I
(Aestivation type)

- Valvate
- Vexillary
- Imbricate
- Twisted

List-II
(Example)

- Cotton
- Pea
- Calotropis*
- Cassia*

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-III, C-IV, D-II
- (4) A-II, B-I, C-III, D-IV

115. Identify the number of plants that show a variation in the length of filaments within a flower and choose the **correct** option.

Salvia, China rose, pea, mustard, lily, citrus

- (1) One (2) Two
- (3) Three (4) Four

116. Match List-I with List-II:

List-I

- Microinjection
- Biolistic
- Restriction endonuclease
- Annealing

List-II

- PCR
- Animal cell
- Plant cell
- Hind II*

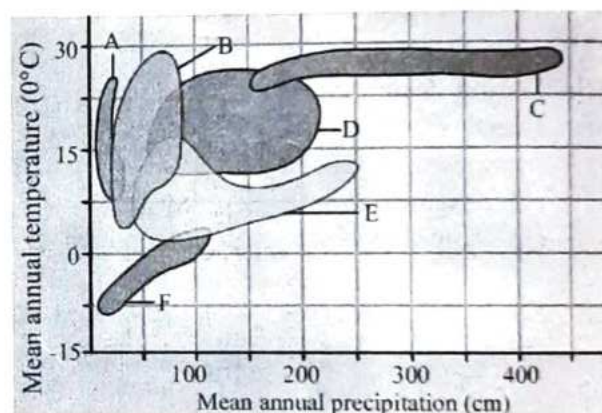
Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-II, B-IV, C-III, D-I

117. Autogamy is:

- (1) functionally self-pollination and genetically cross-pollination.
- (2) genetically self-pollination and functionally cross-pollination.
- (3) cross-pollination both genetically and functionally.
- (4) self-pollination both genetically and functionally.

118. Identify the biomes.



Temperate forests	Grasslands	Arctic and Alpine tundra
(1) D	B	F
(2) C	E	A
(3) E	B	F
(4) F	A	C

119. Match List-I with List-II.

List-I		List-II	
A.	Cytokinin	I.	Stimulates closure of stomata
B.	Ethylene	II.	Increase stem length
C.	Gibberellin	III.	Promotes lateral shoot growth
D.	Abscissic acid	IV.	Synthesised in large amounts by tissues undergoing senescence

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-Issss
 (2) A-II, B-III, C-I, D-IV
 (3) A-I, B-IV, C-II, D-III
 (4) A-III, B-II, C-IV, D-I

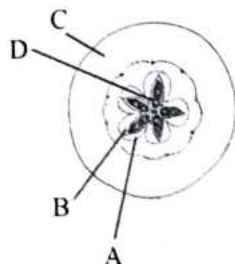
120. In the half-leaf experiment of photosynthesis, KOH solution is used because:

- (1) it provides O_2 to the leaf.
 (2) it provides moisture to the leaf.
 (3) it helps in CO_2 fixation.
 (4) it absorbs CO_2 .

121. Select the **incorrect** statement:

- (1) Growth, differentiation and development are very closely related events in the life of a plant.
 (2) Differentiation is considered as the sum of growth and development.
 (3) Development in plants is under the control of intrinsic and extrinsic factors.
 (4) The intrinsic factor includes both intracellular (genetic) or intercellular factors (chemicals such as plant growth regulators) while the extrinsic factor includes light, temperature, water, oxygen, nutrition, etc.

122. Which part of the fruit, labelled in the given figure makes it a false fruit?



- (1) D → Seed (2) A → Mesocarp
 (3) B → Endocarp (4) C → Thalamus

123. Choose the **mismatched** pair w.r.t. to the phases of cell cycle.

- (1) G_1 phase – Interval between mitosis and initiation of DNA replication.
 (2) G_2 phase – Proteins required for mitosis are synthesised.
 (3) G_0 phase – Cells can proliferate but are metabolically inactive.
 (4) S phase – The amount of DNA per cell doubles.

124. Find the **incorrect** statement w.r.t. the pteridophytes.

- (1) The pteridophytes include horsetails and ferns.
 (2) Pteridophytes are used for medicinal purposes and as soil-binders.
 (3) They are frequently grown as ornamentals.
 (4) Pteridophytes are found in hot and dry places.

125. *Neurospora*, which is used extensively in genetic work, belongs to the class;

- (1) basidiomycetes. (2) ascomycetes.
 (3) deuteromycetes. (4) phycomycetes.

126. Match the List-I with List-II.

List-I		List-II	
A.	Prophase II	I.	Marks the end of meiosis
B.	Metaphase II	II.	Initiated immediately after cytokinesis
C.	Anaphase II	III.	Chromosomes align at the equator in this stage
D.	Telophase II	IV.	Begins with the simultaneous splitting of the centromere

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 (2) A-II, B-III, C-IV, D-I
 (3) A-I, B-IV, C-III, D-II
 (4) A-IV, B-II, C-III, D-I

127. Pachytene is characterized by the:
- appearance of recombination nodules.
 - terminalization of chiasmata.
 - compaction of chromosomes.
 - chromosomes alignment on equatorial plate.
128. The enzyme involved in alcoholic fermentation of pyruvate is:
- citrate synthase.
 - lactate dehydrogenase.
 - alcohol dehydrogenase.
 - succinate dehydrogenase.
129. Most prokaryotic cells, particularly the bacterial cells have:
- a chemically simple cell envelope.
 - a chemically complex cell envelope.
 - no cell envelope.
 - cell envelope only in the form of a cell membrane.
130. In 2005, for each of the 14 million people present in a country, 0.028 were born and 0.008 died during the year.
Using exponential equation, the number of people present in 2015 is predicted as:
- 25 millions
 - 17 millions
 - 20 millions
 - 18 millions
131. Identify the **correct** statement(s) w.r.t. the G_0 phase of cell cycle.
- It is called the quiescent stage of the cell cycle.
 - Cells in this stage remain metabolically active.
 - Cells in this stage no longer proliferate unless called on to do so depending on the requirement of the organism.
- A and B only.
 - A only.
 - B and C only.
 - A, B and C.
132. Given below are two statements:
- Statement-I:** In plants, the ability to follow different pathways in response to environment or phases of life to form different kinds of structures is called plasticity.

Statement-II: In coriander and larkspur plant, the leaves of the juvenile plant are different in shape from those in mature plants.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

133. Match list I with list II.

List-I (Various types of cell and organism)	List-II (Size)
A. Typical bacteria	I. 10-20 μm
B. Viruses	II. 1-2 μm
C. PPLO	III. 0.1 μm
D. A typical eukaryotic cell	IV. 0.02-0.2 μm

Choose the **correct** answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-II, B-IV, C-III, D-I
- A-III, B-IV, C-I, D-II
- A-IV, B-II, C-I, D-III

134. The Central Dogma of molecular biology states that, genetic information flows from:

- DNA $\xrightarrow{\text{Transcription}}$ mRNA $\xrightarrow{\text{Translation}}$ Protein.
- mRNA $\xrightarrow{\text{Transcription}}$ DNA $\xrightarrow{\text{Translation}}$ Protein.
- Protein $\xrightarrow{\text{Transcription}}$ DNA $\xrightarrow{\text{Translation}}$ mRNA
- Protein $\xrightarrow{\text{Transcription}}$ mRNA $\xrightarrow{\text{Translation}}$ DNA

135. Karyokinesis not followed by cytokinesis leads to the formation of:

- uninucleated cell.
- multinucleated cell.
- cell with no nucleus.
- cell with no cytoplasm.

Botany : Section-B (Q. No. 136 to 150)

136. Arrange the steps involved in developing a transgenic plant resistant to nematodes using RNA interference:

- Production of both sense and anti-sense RNA in the host cells.
- Silencing of the specific mRNA of the nematode, leading to the disruption of its survival.
- Introduction of nematode-specific genes into the host plant using *Agrobacterium* vectors.
- Formation of double-stranded RNA (dsRNA) in the host cells.
- Protection of the transgenic plant from the nematode.

Choose the correct sequence from the options given below:

- (c), (a), (d), (b), (e)
- (b), (a), (c), (d), (e)
- (b), (a), (d), (c), (e)
- (a), (c), (b), (e), (d)

137. Given below are two statements:

Statement-I: The synergids have special cellular thickenings at the micropylar tip called filiform apparatus in angiosperms.

Statement-II: Egg apparatus consists of one synergid and one egg cell.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

138. Following are the main parts of a dorsiventral leaf:

- Spongy parenchyma
- Palisade parenchyma
- Abaxial epidermis
- Adaxial epidermis
- Cuticle

Choose the correct sequence of parts (upper to lower side) from the options given below:

- (e), (d), (c), (b), (a)
- (e), (d), (b), (a), (c)
- (e), (c), (b), (a), (d)
- (e), (c), (a), (b), (d)

139. Consider the following components of the plant cell wall:

- Secondary wall
- Plasma membrane
- Primary wall
- Middle lamella
- Cytoplasm

Arrange these components in the sequence in which they are positioned in the plant cell from outside to inside:

- (e)-(c)-(a)-(d)-(b)
- (c)-(a)-(d)-(b)-(e)
- (d)-(c)-(a)-(b)-(e)
- (c)-(d)-(e)-(a)-(b)

140. The discontinuously synthesised fragment in DNA replication are later joined by the enzyme;

- DNA ligase.
- DNA dependent DNA polymerase.
- permease.
- DNA dependent RNA polymerase.

141. Match List-I with List-II.

List-I	List-II
A. <i>Panthera pardus</i>	I. Carnivora
B. <i>Mangifera indica</i>	II. Poales
C. <i>Musca domestica</i>	III. Sapindales
D. <i>Triticum aestivum</i>	IV. Diptera

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-II, C-IV, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-I, B-IV, C-III, D-II

142. The two antibiotic resistance genes on vector pBR322 are for:

- (1) ampicillin and tetracycline.
- (2) ampicillin and chloramphenicol.
- (3) chloramphenicol and tetracycline.
- (4) tetracycline and kanamycin.

143. Read the following statements related to the goals of human genome project.

- A. Improve tools for data analysis.
- B. Store information in databases.
- C. Address the ethical, legal, and social issues that may arise from the project.
- D. Transfer related technologies to other sectors, such as industries.

Choose the *most appropriate* answer from the options given below:

- (1) A and B only (2) A only
- (3) B and C only (4) A, B, C and D

144. The process of removal of anther from the flower bud before it dehisces is called as;

- (1) emasculation.
- (2) bagging.
- (3) artificial hybridisation.
- (4) pollen pistil interaction.

145. Which of the following components plays a role in bundle sheath cells during Hatch and Slack pathway?

- (a) RuBisCO
- (b) PEPcase
- (c) Phosphoenolpyruvate (PEP)
- (d) Malic acid
- (e) CO₂

Choose the correct answer from the options given below:

- (1) (a), (b), and (d) only
- (2) (b), (c), (d), and (e) only
- (3) (a), (c), and (e) only
- (4) (a), (d) and (e) only

146. Find out incorrect statement for cross between homozygous tall and homozygous dwarf in pea plant.

- (1) F₁ generation is tall.
- (2) Tt expresses dominant character.
- (3) TT represents heterozygous condition.
- (4) Dwarf progeny was obtained in F₂ generation after selfing F₁.

147. Select the incorrect statement.

- (1) The base of each ecological pyramid represents the producers or the first trophic level while the apex represents tertiary or top-level consumer.
- (2) A given organism can occupy only one trophic level.
- (3) The trophic level represents a functional level, not a species as such.
- (4) Energy at a lower trophic level is always more than at a higher level.

148. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): M phase is also called the equational division.

Reason (R): The number of chromosomes in the parent cells and progeny cells formed after M phase is the same.



In the light of the above statements, choose the correct answer from the options given below:

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

149. Match the following column-I with column-II and choose the correct option:

Column-I		Column-II	
(a)	First decarboxylation product	(i)	Citrate
(b)	Second decarboxylation product	(ii)	Succinyl CoA $\rightarrow$ Succinic acid
(c)	Substrate level phosphorylation	(iii)	Isocitrate $\rightarrow$ α -ketoglutarate
(d)	Product of condensation of acetyl CoA with oxaloacetate	(iv)	α -ketoglutarate $\rightarrow$ Succinyl CoA

- (1) (a)-(i) (b)-(ii) (c)-(iii) (d)-(iv)
- (2) (a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)
- (3) (a)-(iii) (b)-(iv) (c)-(ii) (d)-(i)
- (4) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)

150. Match List-I with List-II.

List-I		List-II	
A.	Dicotyledonous Root	I.	Sclerenchymatous hypodermis
B.	Monocotyledonous Root	II.	Collenchymatous hypodermis

- | | | | |
|----|-----------------------|------|--|
| C. | Dicotyledonous Stem | III. | More than six xylem bundles & pith is well developed |
| D. | Monocotyledonous Stem | IV. | The pith is small or inconspicuous |

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-I, C-III, D-IV

Zoology : Section-A (Q. No. 151 to 185)

151. Which of the following is the vasoconstrictor?

- (1) ANF
- (2) Renin
- (3) Angiotensin-II
- (4) Histamine

152. Choose the **incorrect** match.

- (1) Coelenterates – Radial symmetry
- (2) Arthropods – Bilateral symmetry
- (3) Platyhelminthes – Triploblastic
- (4) Ctenophores – Triploblastic

153. Match List-I with List-II w.r.t human sperm:

List-I		List-II	
(A)	Head	(I)	Enzymes
(B)	Middle piece	(II)	Sperm motility
(C)	Acrosome	(III)	Energy
(D)	Tail	(IV)	Genetic material

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-I, C-II, D-IV

154. Given below are two statements:

Statement I: In a polysaccharide chain, the left end is called the reducing end.

Statement II: Collagen is the most abundant protein in animal world.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

155. Read the following statements (A-E):

- A. MTP is to get rid of unwanted pregnancies due to casual unprotected intercourse.
- B. Second trimester abortions are much more riskier.
- C. Contraceptives are regular requirements for the maintenance of reproductive health.
- D. Two medical practitioners are required for MTP if the pregnancy is more than 12 weeks.
- E. Saheli has very few side effects and high contraceptive value.

Choose the option with the **correct** statements from the options given below:

- (1) B, D and E only
- (2) A, B, C and D only
- (3) A, B, D and E only
- (4) A, C and E only

156. Match List-I with List-II:

List-I	List-II
(A) First transgenic cow	(I) mRNA silencing
(B) Safety test of polio vaccine	(II) Golden rice
(C) Resistance from nematode	(III) Rosie
(D) Vitamin 'A' enriched crop	(IV) Transgenic mice

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-IV, C-I, D-II

157. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Amphibians like frogs can respire through their moist skin.

Reason R: Respiration that occurs by skin is known as cutaneous respiration.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

158. Globular regulatory protein of skeletal muscle fibres is:

- (1) tropomyosin. (2) troponin.
- (3) actin. (4) myosin.

159. Ball and socket joint is found in:

- (1) shoulder joint.
- (2) hip joint.
- (3) between atlas and axis.
- (4) Both (1) and (2)

160. The exoskeleton of arthropods is made up of:

- (1) inulin. (2) chitin.
- (3) cellulose. (4) collagen.

161. Which of the following are not the effects of Parathyroid hormone?

- A. Stimulates the process of bone resorption
- B. Decrease Ca^{2+} level in blood
- C. Reabsorption of Ca^{2+} by renal tubules

- D. Decrease the absorption of Ca^{2+} from digested food
 E. Increases metabolism of carbohydrates
 Choose the most appropriate answer from the options given below:

- (1) B and C only (2) A and C only
 (3) B, D and E only (4) A and E only

162. Match List-I with List-II:

List-I	List-II
(A) Typhoid	(I) <i>Plasmodium</i>
(B) Pneumonia	(II) <i>Streptococcus pneumoniae</i>
(C) Malaria	(III) <i>Salmonella typhi</i>
(D) Ringworm	(IV) <i>Microsporium</i>

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-III, B-I, C-II, D-IV
 (4) A-IV, B-II, C-I, D-III

163. Given below are two statements:

Statement I: Erythrocytes, leucocytes and platelets are collectively called formed elements.

Statement II: Calcium ions play a very important role in clotting.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

164. Choose the **correct** genotype of sex chromosome present in this female bird.

- (1) XX (2) XO
 (3) ZW (4) ZZ

165. Given below are two statements.

Statement I: The son of a woman who carries the gene for colour blindness has a 50% chance of being colour blind.

Statement II: Colour blindness is an autosomal recessive trait.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

166. Complete the analogy w.r.t. innate immunity.

Interferons: Cytokine barriers :: Mucus coating of epithelial lining: _____

- (1) Physical barriers
 (2) Physiological barriers
 (3) Cellular barriers
 (4) Blood brain barriers

167. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The skeleton of sponges is made up of spicules.

Reason R: Spongin fibres cannot support the body.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is correct but R is not correct.
 (2) A is not correct but R is correct.
 (3) Both A and R are correct and R is the correct explanation of A.
 (4) Both A and R are correct but R is NOT the correct explanation of A.

168. The functional unit of the contractile system in the striped muscle is:

- (1) I-band. (2) A-band.
 (3) sarcomere. (4) myofibril.

169. Which among these is the **correct** combination of aquatic mammals?

- (1) Dolphins and *Trygon*
- (2) Whales and Dolphins
- (3) *Trygon* and Whales
- (4) Dolphins and Sharks

170. The given diagram represents:



- (1) tubectomy. (2) vasectomy.
- (3) barrier method. (4) implants.

171. Frogs differ from humans in possessing:

- (1) hepatic portal system.
- (2) paired cerebral hemispheres.
- (3) nucleated red blood cells.
- (4) thyroid as well as parathyroid.

172. Which of the following are primary lymphoid organs?

- (1) Bone marrow and tonsils
- (2) Lymph nodes and thymus
- (3) Bone marrow and Peyer's patches
- (4) Thymus and bone marrow

173. Given below are two statements:

Statement I: The neural system is better organised in insects, where a brain is present along with a number of ganglia and neural tissues.

Statement II: The CNS is divided into two divisions called somatic neural system and autonomic neural system.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

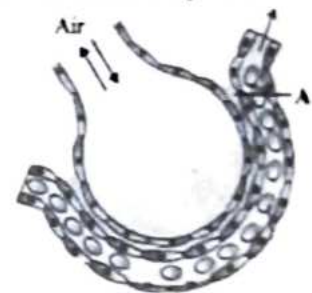
174. Match List-I with List-II:

List-I	List-II
(A) Jointed appendages	(I) <i>Asterias</i>
(B) Water vascular system	(II) <i>Culex</i>
(C) Canal system	(III) <i>Spongilla</i>
(D) Flame cells	(IV) <i>Taenia</i>

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-I, B-III, C-II, D-IV
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-I, C-III, D-IV

175. Identify the part labelled as **A** in the given diagram and select the **correct** option.



- (1) Basement membrane
- (2) Blood vessel
- (3) Pleura
- (4) Compound epithelium

176. Find the **mismatched** pair.

- (1) Order Primata - Human
- (2) Order Carnivora - Cat
- (3) Phylum Chordata - Housefly
- (4) Order Polymoniales - *Petunia*

177. Match List-I with List-II:

List-I		List-II	
(A)	Diaphragms	(I)	Inhibit ovulation
(B)	Contraceptive pills	(II)	Decrease fertilising capacity of sperm.
(C)	Cu releasing IUDs	(III)	Absence of menstrual cycle and ovulation following parturition.
(D)	Lactational amenorrhea	(IV)	Cover the cervix blocking the entry of sperms.

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
 (2) A-IV, B-I, C-II, D-III
 (3) A-II, B-IV, C-I, D-III
 (4) A-III, B-II, C-I, D-IV

178. A haemophilic woman (A) marries normal man (B). This couple has one daughter (C) and one son (D). After sometime this son (D) marries with a normal girl (E) whose father was having haemophilia.

Which of the following is **correct** about offsprings of (D) & (E)?

- (1) 50% daughters are haemophilic while all the sons are normal.
 (2) All the daughters are haemophilic while 50% sons are haemophilic.
 (3) 50% daughters are haemophilic while 50% sons are haemophilic.
 (4) 50% daughters normal and carrier while all the sons are haemophilic.

179. Given below are two statements:

Statement I: In phylum Ctenophora, reproduction takes place only by sexual means.

Statement II: The water canal system is the most distinctive feature of echinoderms.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

180. Which of the following type of tissue is being described by the given statements?

- A. They secrete modified polysaccharides, which accumulate between cells and fibres and act as matrix.
 B. They include cartilage, bone, adipose and blood.
 C. They provide strength, elasticity and flexibility to the tissue.

- (1) Epithelial tissue
 (2) Connective tissue
 (3) Muscle tissue
 (4) Neural tissue

181. Natural selection, where more individuals acquired specific character value other than the mean character value at both ends of the distribution curve, leads to:

- (1) random change. (2) stabilising change.
 (3) directional change. (4) disruptive change.

182. Match the columns and choose the appropriate option.

Column-I		Column-II	
(a)	Rete testis	(i)	Spermatogonia
(b)	Urethral meatus	(ii)	Accessory duct
(c)	Male germ cell	(iii)	External opening
(d)	Seminal vesicle	(iv)	Accessory gland

- (1) (a)-(ii) (b)-(iii) (c)-(i) (d)-(iv)
 (2) (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)
 (3) (a)-(i) (b)-(iii) (c)-(iv) (d)-(ii)
 (4) (a)-(iii) (b)-(ii) (c)-(i) (d)-(iv)

183. Match List-I with List-II:

List-I		List-II	
(A) Thymine	(I)	Substituted pyrimidine	
(B) Guanosine	(II)	Sugar found in RNA	
(C) Ribose	(III)	Nucleoside	
(D) Abrin	(IV)	Toxin	

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-IV, B-II, C-I, D-III
 (4) A-III, B-I, C-II, D-IV

184. How many secondary spermatocytes are required to form 400 million spermatozoa?

- (1) 400 million (2) 50 million
 (3) 100 million (4) 200 million

185. How many O_2 molecules can bind with single molecule of Hb?

- (1) 8 (2) 6
 (3) 4 (4) 5

Zoology : Section-B (Q. No. 186 to 200)

186. Match List-I with List-II:

List-I		List-II	
(A) Loose connective tissue	(I)	Tendon and ligament	
(B) Dense regular connective tissue	(II)	Skin	
(C) Dense irregular connective	(III)	Cartilage and bone	
(D) Specialised connective tissue	(IV)	Macrophages and mast cells	

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
 (2) A-I, B-IV, C-III, D-II
 (3) A-IV, B-I, C-II, D-III
 (4) A-IV, B-II, C-I, D-III

187. Given below are two statements:

Statement I: Infectious diseases are very rare.

Statement II: *Plasmodium* enters the human body as sporozoites.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

188. Kidneys are situated between the levels of

- (1) last thoracic and 3rd lumbar vertebra
 (2) 1st thoracic and 1st lumbar vertebra
 (3) 1st thoracic and last lumbar vertebra
 (4) 3rd thoracic and 2nd lumbar vertebra

189. Read the following statements (A-D).

- A. Each protein is a polymer of amino acids and lipids.
 B. Glycine is the simplest amino acid.
 C. Paper made from plant pulp is cellulosic.
 D. N-acetyl glucosamine is a chemically modified sugar.

Choose the option with the **correct** statements from the options given below:

- (1) A and B only
 (2) B and C only
 (3) B, C and D only
 (4) A, B and C only

190. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The chemicals stored in the synaptic vesicles are termed as neurotransmitters.

Reason (R): They can transmit signals from neuron to neuron/other target cell.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

191. Match List-I with List-II:

List-I		List-II	
(A)	Pituitary gland	(I)	Grave's disease
(B)	Thyroid gland	(II)	Diabetes mellitus
(C)	Adrenal cortex	(III)	Diabetes insipidus
(D)	Pancreas	(IV)	Addison's disease

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-III, B-II, C-I, D-IV
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-I, C-IV, D-III

192. What is the osmolarity (in mOsmol L^{-1}) in the outer cortex and inner medulla region?

- (1) 300 and 600, respectively
- (2) 600 and 300, respectively
- (3) 1200 and 300, respectively
- (4) 300 and 1200, respectively

193. Match List-I with List-II:

List-I		List-II	
(A)	Adaptive radiation	(I)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(B)	Convergent evolution	(II)	Bones of forelimbs in man and whale
(C)	Divergent evolution	(III)	Wings of butterfly and bird
(D)	Evolution by anthropogenic action	(IV)	Australian marsupials

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-III, D-II
- (4) A-IV, B-III, C-II, D-I

194. Match List-I with List-II:

List-I		List-II	
(A)	Epiglottis	(I)	Air sacs of lungs
(B)	Alveoli	(II)	Wind-pipe
(C)	Larynx	(III)	Cartilaginous flap
(D)	Trachea	(IV)	Voice box

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-III, B-II, C-IV, D-I
 (3) A-III, B-I, C-IV, D-II
 (4) A-IV, B-I, C-II, D-III
- 195.** Persons with 'AB' blood group are called as "universal recipients". This is due to:
 (1) absence of antigens A and B in plasma.
 (2) presence of antibodies, anti-A and anti-B, on RBCs.
 (3) absence of antibodies, anti-A and anti-B, in plasma.
 (4) absence of antigens A and B on the surface of RBCs.
- 196.** The teeth of the gizzard in cockroaches are composed of:
 (1) calcium. (2) cellulose.
 (3) chitin. (4) None of these
- 197.** Which of the following statement(s) are **incorrect**?
 A. Placenta also acts as an exocrine tissue and produces several hormones.
 B. Inner cell mass differentiates into ectoderm and endoderm just before implantation.
 C. After implantation, finger-like projections appear on the trophoblast called chorionic villi.
 D. During pregnancy, the levels of hormones like cortisol, prolactin, thyroxine, etc., are increased several folds in the fetal blood.
 (1) A only (2) B and D only
 (3) A, B and D only (4) C and D only

198. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask.

- (1) CH_4 , H_2 , NH_3 and water vapor at 800°C .
 (2) CH_4 , H_2 , NH_3 and water vapor at 600°C .
 (3) CH_4 , O_2 , NH_3 and water vapor at 600°C .
 (4) CH_4 , O_2 , NH_3 and water vapor at 800°C .

199. Sequence of meninges from inside to outside is:

- (1) Dura mater – arachnoid – pia mater
 (2) Dura mater – pia mater – arachnoid
 (3) Arachnoid – dura mater – pia mater
 (4) Pia mater – arachnoid – dura mater

200. Given below are four statements pertaining to separation of DNA fragments using Gel electrophoresis. Identify the **incorrect** statements.

- A. DNA is negatively charged molecule and so it is loaded on gel at the anode terminal.
 B. DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.
 C. Smaller the size of DNA fragment, the farther it travels during gel electrophoresis.
 D. Pure DNA can be visualized directly by exposing to UV radiation.

Choose **correct** answer from the options given below:

- (1) A, C and D only
 (2) A, B and C only
 (3) B, C and D only
 (4) A, B and D only